

BoDemGioiHan	
_id	String
count	Number
firstAt	Date NN
blockedUntil	Date
expiresAt	Date NN

Banner 	
id 	ObjectId
title	String NN
description	String NN
buttonText	String
linkUrl	String
backgroundColor	String
textColor	String
imageUrl	String
cloudinaryId	String
displayType 	String
discountText	String
discountSubtext	String
order	Number
isActive	Boolean
page 	String NN
createdAt	Date
updatedAt	Date

Provider 	
_id 	ObjectId
name	String NN
slug 	String NN
logo	String
type 	String

Category	
_id	ObjectId
name	String NN
slug	String NN
icon	String

User	
id	ObjectId
userId	String
name	String
fullName	String
birthday	Date
email	String
password	String
avatar	String
avatarPublicId	String
passwordChangedAt	Date
role	String
provider	ObjectId
bio	String
phone	String
status	Boolean
googleId	String
emailVerified	Boolean
verifyTokenHash	String
verifyTokenExp	Date
solCoin	Number
createdAt	Date
updatedAt	Date

Course	
id 	ObjectId
title	String NN
slug 	String NN
description	String NN
thumbnail	String
price 	Number NN
provider 	ObjectId
level 	String
instructor 	ObjectId NN
totalDuration	Number
rating	Number
reviewsCount	Number
studentsCount	Number
isPublished 	Boolean
isPopular	Boolean
isTrending	Boolean
isNewRelease	Boolean
createdAt	Date
updatedAt	Date

Post 📄	
id 🔗	ObjectId
title	String NN
slug 🏷️	String NN
excerpt	String NN
content	String NN
thumbnail	String
topic 📁	String
tags	Array
author 🔗	ObjectId NN
views	Number
isPublished	Boolean
createdAt	Date
updatedAt	Date

Document 	
id 	ObjectId
title	String  
description	String  
fileUrl	String  
filePublicId 	String
fileName	String  
fileExt 	String  
fileSize	Number  
uploader 	ObjectId 
downloadCount	Number
createdAt	Date
updatedAt	Date

Enrollment	
id	ObjectId
course	ObjectId NN
student	ObjectId NN
totalProgress	Number
status	String
completedAt	Date
lastAccessedAt	Date
finalScore	Number
createdAt	Date
updatedAt	Date

Certificate 	
id 	objectId
course 	objectId NN
student 	objectId NN
certificateNumber 	String Max
title	String NN
description	String
completionDate	Date NN
courseName	String
instructorName	String
courseDuration	String
finalScore	Number
scorePercentage	Number
isPublic	Boolean
verificationUrl	String
verificationCode 	String
issuedAt	Date
expiresAt 	Date
isValid	Boolean
signedBy	String
createdAt	Date
updatedAt	Date

Lesson	
_id	ObjectId
courseId	ObjectId NN
title	String NN
slug	String NN
content	String
videoUrl	String
documentUrl	String
duration	String
order	Number
createdAt	Date
updatedAt	Date

FAQ	
_id	ObjectId
courseId	ObjectId
question	String NN
answer	String NN
createdAt	Date
updatedAt	Date

Course_Category	
course	ObjectId NN
category	ObjectId NN

Review	
_id	ObjectId
course	ObjectId NN
student	ObjectId NN
rating	Number NN
comment	String NN
helpful	Number
unhelpful	Number
isVerifiedPurchase	Boolean
createdAt	Date
updatedAt	Date

Order ID	
id	ObjectId
code	String NN
course	ObjectId NN
student	ObjectId NN
amount	Number NN
status	String
expiresAt	Date NN
paidAt	Date
confirmedBy	ObjectId
note	String
daBaoChuyenKhoanLuc	Date
createdAt	Date
updatedAt	Date

GiaoDichCoin	
_id	ObjectId
hocVien	ObjectId NN
loai	String NN
soCoin	Number NN
soDuSau	Number NN
khoa	ObjectId
ghiChu	String
nguoiTao	ObjectId
createdAt	Date
updatedAt	Date

Enrollment_lessonProgress	
enrollment	ObjectId NN
lesson	ObjectId NN
status	String
watchedDuration	Number
completedAt	Date

Quiz	
id	ObjectId
course	ObjectId
lesson	ObjectId
title	String
description	String
passingScore	Number
timeLimit	Number
attempts	Number
randomizeQuestions	Boolean
randomizeOptions	Boolean
showAnswers	Boolean
totalPoints	Number
isPublished	Boolean
createdAt	Date
updatedAt	Date

<u>id</u>	ObjectId
user	ObjectId NN
course	ObjectId NN
lesson	ObjectId
createdAt	Date
updatedAt	Date

Quiz_questions	
_id	ObjectId
quiz	ObjectId NN
text	String NN
type	String
options	Array
correctAnswer	String
explanation	String
points	Number

QuizAttempt	
_id	ObjectId
quiz	ObjectId NN
student	ObjectId NN
score	Number NN
percentage	Number NN
passed	Boolean NN
timeSpent	Number
startedAt	Date
submittedAt	Date NN
attemptNumber	Number
retryReason	String
feedback	String
status	String
createdAt	Date
updatedAt	Date

Achievement 	
_id 	ObjectId
student 	ObjectId 
type 	String 
title	String 
description	String 
badgeImage	String
relatedCourse 	ObjectId
relatedQuiz 	ObjectId
level 	String
points	Number
isPublic	Boolean
unlockedAt	Date
createdAt	Date
updatedAt	Date

CuocTroChuyen_tinNhan	
cuocTroChuyen	ObjectId NN
valTro	String NN
noiDung	String NN
createdAt	Date

QuizAttempt_answers	
attempt	ObjectId N
questionId	ObjectId N
studentAnswer	Mixed
isCorrect	Boolean
pointsEarned	Number